

TrainCafé: VR Usage on the Go through Virtual Metaphors

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ABSTRACT

As virtual reality becomes increasingly embedded in everyday life, users are adopting it in mobile contexts such as travel. However, immersing in VR introduces challenges for navigating the physical world. While video see-through or removing the headset allows access to real-world navigational information, it also breaks immersion. We present TrainCafé, a prototype that employs virtual metaphors aligned with a real train environment to convey spatial information without disrupting immersive experiences. In an empirical study with 18 participants, our design enabled users to navigate the physical space while remaining fully immersed in VR, while also enhancing engagement and enjoyment. We thus contribute a reality-aware system that blends functional navigation cues with engaging narrative design, and discuss its implications for always-on mobile VR that becomes part of our everyday life.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Interactive systems and tools**.

KEYWORDS

Virtual Reality, Mobile VR, VR on the Go

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1 INTRODUCTION

Travelling by public transport occupies a substantial part of human lives and can be used much more meaningfully and experientially. Research shows that many people perceive travel time as a negative experience, often describing it as boring or stressful [6]. A large proportion choose to spend these journeys reading, working, watching videos or playing games [8, 22].

Mixed reality technologies, now lighter and equipped with integrated video see-through capabilities, are becoming increasingly mobile, enabling “on-the-go” and “in-the-wild” use and being embedded into everyday life, for example during navigation in public spaces [3]. Devices such as the Apple Vision Pro [14] and Meta

Quest 3 [30], already equipped with “travel mode” [29], demonstrate the potential for use within moving vehicles such as planes, trains and cars. They offer immersive virtual spaces for entertainment, work and other activities, that are free from real-world interruptions, and can significantly transform the travel experience. For example, Li et al. [18] showed that embedding a beach-themed VR environment within a vehicle produced a restorative effect on travellers. In addition, Bajorunaite et al. [3] deployed VR headsets “in the wild” and reported a favourable response to VR use on public transport, particularly among passengers travelling alone.

Although travelling with VR in the wild promises rich benefits, challenges remain in physical locomotion within vehicles as well as disconnection from the physical environment. Current solutions, such as embedding see-through portals in the virtual scene [3, 42], switching to a full video see-through mode, or removing the headset altogether, are intrusive, disrupt the narrative flow and break immersion. A promising way is to employ virtual metaphors for conveying physical-world information instead of abruptly exposing the real world. Concepts such as *Substitutional Reality* proposed by Simeone et al. [37] show how physical objects can be paired with virtual counterparts so that spatial layout and affordances are communicated without disrupting immersion. Such metaphor-based approaches can provide a seamless, non-intrusive bridge between virtual and physical spaces, preserving both safety and narrative flow. However, these prior works on immersive navigation and metaphor-based representations have largely been situated in exploratory, abstract, or game-like environments. In contrast, everyday real-world contexts, such as boarding vehicles, navigating narrow aisles, or finding seats, often involve complex spatial constraints and remain largely unexplored.

To address this gap, our work investigates how virtual metaphors can support physical navigation inside a train while users remain fully immersed in VR. We take a first step toward real-world deployment by testing within a simulated, in-the-wild-inspired context to explore locomotion and user experience. While fully in-the-wild studies pose safety challenges, our exploratory approach offers initial insights for designing safe and situated embodied VR experiences. We ask the following research questions:

- RQ1:** Can virtual metaphors effectively help users navigate within a vehicle environment and access real-world information without breaking immersion in VR?
- RQ2:** How do VR metaphors and environments influence users’ overall train travel experiences, compared with the monotonous train interior, especially in terms of engagement?

We propose TrainCafé, a mobile VR experience that maps the interior affordances of a train carriage onto a virtual café, providing physical spatial layout cues for in-train locomotion and embedding diegetic in-VR displays that seamlessly present real-world transit

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and station information. By remapping a constrained train environment through virtual metaphors, we aim to make the space feel larger and more atmospherically engaging, while enhancing safety for navigation in mobile contexts without breaking immersion. We conducted a user study with 18 participants. All participants were able to board the train, navigate the carriage, find their seats, access real-world station information and disembark successfully. Our results show that virtual metaphors can effectively support navigation, and that the café environment was perceived as more enjoyable and engaging than the standard train interior.

Our study thus suggests three key design implications for future in-vehicle VR experiences: first, virtual metaphors not only keep passengers safely oriented while moving through vehicles but also preserve constant immersion. Second, they fit naturally with the trend toward pervasive MR use in the wild, making it viable for designers to consider always-on immersive experiences on the move. Third, in this future, the train carriage could become a canvas for dynamic, adaptive VR spaces, allowing personal choices between a productive virtual office, a classic café or a restorative beach, thus transforming monotonous interiors into engaging and meaningful experiences.

2 RELATED WORK

2.1 VR Usage on the Go

The idea of embedding mixed reality headsets into everyday life and using them in on-the-go scenarios has been studied by the Human-Computer Interaction (HCI) community, especially following recent advances in video see-through technologies. Sun et al. [39] lived with an MR headset for two months, having constant usage across various everyday scenarios, including travelling and commuting, where MR created an immersive space that fulfil diverse user needs. Research has highlighted a range of benefits associated with using VR/XR in vehicular contexts. For instance, Yavo-Ayalon et al. [44] developed a communal extended-reality system for bus travellers that transforms a mundane ride into a shared social experience. In a similar direction, Lu et al. [20] explored how deliberately diverging physical motion in VR can enhance the entertaining experiences. Beyond leisure applications, mixed reality in cars can also serve as an experimental testbed for prototyping and evaluating new interaction concepts [7]. As vehicle automation increasingly frees passengers from driving, researchers have shown how in-car VR can repurpose the confined cabin into a restorative natural environment [18] or a focused virtual workspace [17]. Such transformations offer promising ways to enrich travel time and even improve productivity through working in VR [33].

However, blocking out reality in vehicles has raised concerns, such as reduced real-world awareness and loss of situational information [2, 23], safety risks like colliding with other people or objects [2, 26], and feelings of loneliness or isolation [23, 39]. In addition, passengers often worry about missing destination stops or important travel announcements [2, 43]. To address these issues, various studies have explored ways to blend real-world information into mobile virtual reality. Examples include using see-through portals to integrate real-world elements such as people or personal devices [42], reconstructing and layering the physical environment within VR [12], incorporating co-located passengers and objects

from vehicles through semi-transparent visualisations, and enhancing real-world awareness with non-diegetic notifications, spatial markers, or see-through functions [24]. However, these approaches to displaying real-world information are often abrupt, breaking immersion and disrupting the overall experience, rather than integrating seamlessly in a way that matches the tone or narrative of the virtual environment. Virtual metaphors, therefore, may offer a more coherent means of integration. Below, we elaborate on related work in this area.

2.2 Immersive Virtual Metaphors

The presentation of physical objects or real-world elements in VR can be designed in a more natural and metaphorical manner. One major contribution in this area comes from Simeone et al. [37], who introduced the concept of substitutional reality - transforming physical objects such as tables and chairs into interactive affordances within VR. This approach allows users to intuitively understand their functions (e.g., sitting or placing objects) while maintaining narrative coherence by preserving their location and size but altering their virtual appearance. Similarly, Tao et al. [40] blended real-world distractions into VR experiences, for instance, representing the physical wind from a fan as a gently swaying curtain, or visualising the warmth from a heater as glowing fires, thereby conveying real-world subtle distractions in a form that harmonises with the virtual narrative. Metaphors for interactions and abstract information have also been explored, for example, metaphorical door designs were used to represent see-through portals when a bystander approaches [32]. By opening the virtual door, users can momentarily view the real world, enhancing awareness while preserving a sense of control. Li et al. [18] used dynamic virtual elements, such as flying seagulls and an offshore ship, to metaphorically depict vehicle speed and journey progression. For metaphors conveying social information, Medeiros et al. [28] represented nearby bystanders through in-game ghost avatars, and likewise, Li et al. [19] visualised a bystander's communicative intent through the rotation of a virtual avatar. These studies collectively illustrate how metaphorical mappings can seamlessly integrate real-world phenomena into virtual environments to enhance immersion, awareness, and narrative coherence.

3 PROTOTYPE

We chose the train among other modes of transportation because its journey duration is long enough to allow meaningful engagement, yet frequent and accessible enough for everyday use, compared to a bus or a flight. Nonetheless, we regard the concept of VR/MR on the go, realised through metaphorical design, as well as the insights from our exploratory study, as generalisable to other modes of transportation. For the train context, we developed a prototype called TrainCafé, a café-like virtual environment mapped onto the physical layout of a train. This design aims to create a more engaging and pleasant experience than the standard train interior. For safety reasons (see subsection 4.1), we conducted the study in a controlled simulated environment, where we also built a real-train simulation as the baseline condition. The following section elaborates on both prototypes, the simulated real train and the TrainCafé environment. Both prototypes were developed in Unity using the

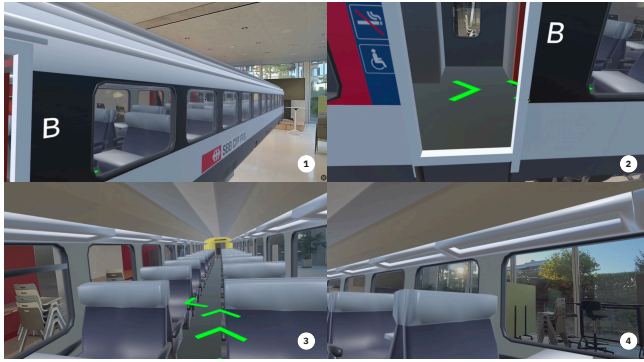


Figure 1: 1) The SBB train is positioned within a spacious physical environment; 2) at the entry of the train, users board the train and follow the green markers indicating the in-train navigation path; 3) green arrow markers guide participants to their designated seat and a diegetic display at the end of the carriage shows real-world travel information; and 4) the train environment is mixed reality, allowing users to see through the train windows and observe the physical reality, resembling the real-world experience of looking out of a train window to observe the world outside.

Meta XR SDK [31], and their source code has been made publicly available on GitHub¹.

3.1 Prototype 1: Train Simulation (Baseline)

A realistic SBB train model was sourced from Sketchfab [27] and implemented at real-world scale, representing a traditional and monotonous train interior without any metaphorical or experiential enhancements (see Figure 1). Two diegetic displays were placed at both ends of the carriage to show the current station and estimated arrival time to Zurich. These displays were mocked using a Wizard-of-Oz approach and were not connected to live data via the SBB API [35]. The train carriage can be positioned at arbitrary locations and orientations within the physical environment. A physical chair was placed to precisely match the designated virtual seat. To guide participants, green arrow markers were displayed in the virtual environment, indicating the correct seat position.

3.2 Prototype 2: TrainCafé

To increase enjoyment and preserve immersion while still providing real-world awareness such as the physical spatial layout and travel information, we developed a virtual Café environment mapped onto the same train carriage model (see Figure 2). This environment integrates diegetic cues that connect to the real-world travel context, such as the station and estimated arrival time. The design employs several virtual metaphors: stanchions indicate the boundaries and width of the aisle, a sofa represents the physical train seat, and a café door symbolises the connecting areas of the carriage. These elements feel more natural than guardian-system-like virtual walls and preserve the narrative of relaxing in a cosy café. Through such affordances, users can maintain spatial awareness



Figure 2: 1) After users board the train, the interior of the train carriage transforms into a VR café with chairs, tables, a piano, and chess sets; 2) the green arrow markers guide users to the designated seat; the stanchions with red ropes work as natural barriers to indicate spatial information (where the seats and obstacles are and where the aisle is), enabling users to walk while fully immersed in VR; 3) a display shows real-world information including the next station and arrival time; and 4) red arrow markers indicate ways to exit the train.

for navigation while experiencing a larger-than-real environment – the café extends beyond the actual train interior, offering a sense of expanded space. The design could also be adapted into boundless environments, creating a feeling of vastness that aligns with different narrative intentions. Upon entering the train, the system switches to full VR mode, transforming the virtual train setting into a cosy and emotionally engaging café. The same design concept could also be adapted for other purpose-driven environments, such as a virtual office for productive travel.

4 USER STUDY

To examine whether a metaphorical VR environment can enhance the train ride experience, making it more enjoyable and engaging, while still maintaining real-world awareness and supporting foundational travel tasks, we conducted a user study with 18 participants. The study followed a within-group design with two conditions: 1) a baseline condition simulating a real train in mixed reality, and 2) a café condition in virtual reality. The order of the conditions was counterbalanced using a Latin square [25]. This research was approved by our university’s ethics board and all participants gave informed consent prior to their participation.

4.1 Lab Study Reasoning

Although the intended use case of our prototype is VR usage in the wild, conducting the study in a real train environment posed substantial safety challenges. For example, despite efforts to map the VR interior to real train layouts, train models differ dynamically, making consistent spatial alignment difficult. Moreover, the current prototype does not yet handle all complex real-world factors, such as obstacle detection and bystander recognition, which increases the risk of collisions or injury during navigation in VR. Therefore,

¹<https://github.com/Panogu/TrainCafeVR>

we opted for a controlled laboratory study as a first step, simulating train scenarios in a safe setting.

4.2 Study Setup & Procedure

The study was conducted in a spacious room approximating the size of a train carriage. The virtual train was consistently positioned in the same physical location. A physical chair matching the height and placement of the virtual seat was used, and its position is verified before each session. During each session, one researcher observes the whole scenario and participants' behaviour, especially regarding safety. In addition, casting was enabled and streamed to a laptop, allowing observations of participants' activities in VR/MR.

Participants provided informed consent for data recording, completed a demographic questionnaire, and were then directly fitted with the headset. No familiarisation procedure was conducted to prevent learning effects. Participants were instructed that they would begin a journey to Zurich and were asked to complete the following tasks: 1) board the train, 2) follow the green marks in the train to find their seat, 3) sit there and follow travel-related instructions on the display, and 4) disembark from the train after arrival. The instructions remained the same for both conditions. Each task lasted approximately two minutes and was followed by post-condition questionnaires. After completing both conditions, participants took part in a semi-structured interview lasting about ten minutes. In total, the study session lasted around thirty minutes.

4.3 Participants

Participants were recruited through word of mouth, with no compensation. Eighteen participants (4 female, 14 male) took part in the study. They ranged in age from 20 to 44 years ($M = 25.33$, $SD = 7.19$), and all are currently pursuing a bachelor's, master's, or doctoral degree. Regarding VR usage, nine participants reported using VR only a few times a year, while seven participants (39%) stated that they never use VR. Only two participants reported regular use. In terms of train travel experience, most participants reported frequent train use, fourteen participants travelled by train at least once a week, including four participants (22.2%) who used it daily or almost daily. The remaining four participants (22.2%) travelled less regularly, about once a month.

4.4 Measurements & Analysis

We employed the following questionnaires to measure workload, spatial presence and usability: NASA Task Load Index (NASA-TLX) [11] was used to assess participants' perceived workload with 21 gradations on the scales across six dimensions. Spatial Presence Experience Scale (SPES) [13] was employed to measure the participants' sense of "being there". The first ten items were used, as they specifically address the construct of spatial presence. Single Ease Question (SEQ) [34] was used to evaluate the overall perceived usability of the system. Besides, we included self-defined questions to measure the perceived coolness and enjoyability of the experience on a 7-point Likert scale (1 = Strongly disagree, 7 = Strongly agree): 1) The environment felt cool. 2) The experience was fun and enjoyable. In addition, we observed whether participants could successfully navigate, locate their assigned seat, and complete the task.

For the semi-structured interview, participants were asked about 1) their overall impressions of each condition, including what aspects stood out positively and negatively; 2) the alignment between the physical and virtual environments; 3) their experience of navigation and orientation within the train; and 4) their sense of real-world awareness throughout the experience. We transcribed the interview data verbatim and performed thematic analysis [41]. Below, we report our quantitative and qualitative results.

5 RESULTS

Overall, all participants successfully completed the task, physically navigating to the seat while fully in VR/MR. Quantitative results revealed consistently low workload levels across conditions. The overall NASA-TLX scores indicated similar perceived workload in the baseline ($M = 14.89$, $SD = 10.57$) and TrainCafé ($M = 15.48$, $SD = 13.24$) conditions. Usability ratings on the SEQ were comparably high for both (baseline: $M = 6.22$, $SD = 0.65$; TrainCafé: $M = 6.17$, $SD = 0.71$), indicating that both systems were perceived as easy and intuitive to use. Spatial presence was slightly higher in the TrainCafé condition ($M = 3.55$, $SD = 0.83$) compared to the baseline ($M = 3.23$, $SD = 0.88$). Participants also rated the TrainCafé as cooler ($M = 5.83$, $SD = 1.29$) and marginally more enjoyable ($M = 5.44$, $SD = 1.58$) than the baseline condition ($M = 5.17$, $SD = 1.58$; $M = 5.11$, $SD = 1.68$, respectively).

Through thematic analysis, we identified two overarching layers shaping participants' travel experiences: (1) *Navigating the Spaces* (Functional Interaction Layer), which captures how participants used virtual metaphors to orient themselves in physical spaces, perceived affordances in virtual spaces, and built embodied trust across both; (2) *Living the Journey* (Experiential Layer), which describes how participants emotionally, aesthetically, and sensorially engaged with the journey.

5.1 Navigating the Spaces

5.1.1 Spatial Affordances & Access to Real-World Information. In general, the virtual metaphors worked well to indicate the spatial layout. Several participants described the guardrails, ropes, and other cues as "comprehensible," "straightforward," and "clear to follow". One participant suggested that such visual augmentation could also assist real-world navigation by visually circling obstacles with virtual barriers similar to those used at construction sites.

Regarding travel information, participants mentioned that the SBB display felt familiar and realistic, and was convenient for obtaining travel details. However, it risked "*not fitting the vibe of the whole place at all*" (P5) and some suggested that it could be "*integrated in a sneaky way*", for example, "*while reading the menu you could have the next destination [in the menu]*".

Design Insights: 1) Virtual metaphors can effectively communicate spatial information and serve as intuitive affordances in both virtual and physical contexts. 2) Diegetic information displays in VR can be designed more creatively, adapting to the aesthetic and narrative tone of the space rather than being limited to conventional display formats.

5.1.2 Spatial Matching & Mental Models & Trust. Most participants (16/18) reported that the alignment between the physical and virtual chairs was accurate, and the seating process felt smooth. They also emphasised the importance of precise spatial matching for maintaining mental models and a sense of safety, noting that *“only a few centimetres of difference could already be super confusing”* (P11) and that *“a mismatch could cause an insecure feeling and worry about bumping into the chair”* (P1).

Beyond positional alignment, spatial scale alignment also played a role, with misalignments disrupting participants’ mental models and leading some to experience *“surprise”* and *“confusion”*. Upon entering the café, participants described feeling *“weird”* about the sudden change in spatial scale, noting that *“you know the train space is confined, then inside it is a huge space; it is spatial manipulation”* (P5). The unexpected expansion led one participant to step back out of the café to observe the transition again (P12). Other weird behaviours were also observed. Several participants were trying to touch the virtual table, later describing how *“the brain was tricked. It thought the table was there, but actually I was not touching anything”* (P9).

Though spatial information could be communicated effectively through visual affordances, issues of trust still emerged. Several participants expressed that *“you know physically you are somewhere else, you cannot fully trust the system”* (P5), and that *“visual information is not always reliable”* (P3). As a result, some felt *“worried about safety issues because [they were] completely cut off from the real environment”*, and consequently *“felt insecure”*. To compensate, most participants relied on haptic support, using their legs or feet to reach out or their hands to touch and feel the chair before sitting down. This embodied checking behaviour provided reassurance when visual cues alone felt insufficient. However, not all participants shared these concerns: several stated that they *“completely trusted the environment”* (P8, P10, P14). Others described a gradual process of embodied learning, where *“trust can build up”* (P7), and *“if it always works several times, you get used to it and have more confidence”* (P9).

5.1.3 Embodiment & Proprioceptive Cues. Participants also described a sense of weirdness resulting from reduced embodiment during physical navigation, noting that they *“could not see [their] feet”* (P1), and that *“you are walking, but you don’t see your body move”* (P13). They reported a weakened sense of bodily awareness and suggested that *“the presence of a physical body could increase the sense of realism”*.

During the boarding and walking process, several embodied behaviours reflecting proprioceptive confusion were observed. Participants approached the train with exaggerated caution, taking large steps to board even though the physical floor was flat. They described an automatic interpretation of spatial reality: *“In the real world, you have the height difference between the train and the platform. Even though the gap is nonexistent, the brain is still in real-world mode”* (P6). Another participant noted, *“The brain just went off. It’s on the same floor. I could just step through it, but still did weird gestures”* (P4). This perceptual confusion made participants more cautious and, at times, fearful when moving through both the physical and virtual environments. We also observed participants who were *“tempted to hold [their] arms still and not swing them*

around”, while navigating the aisle, and others who felt *“scared and worried about hitting anything”*, even though the surrounding seats were purely virtual. Several participants also instinctively lowered their heads when entering the café, reacting to the presence of a virtual door despite knowing that no physical obstacle existed.

Design Insights: 1) Acknowledge reduced embodiment and protective bodily behaviour as design signals. Designers can intentionally introduce cues that reinforce bodily confidence or avoid potential confusion in environmental design to increase user trust, reduce anxiety, and promote fluid, safe movement. 2) Design for affordance alignment [21] and leverage embodied memory rather than fight it. The ducking behaviour under a virtual doorway reveals how affordances in virtual environments can evoke real-world motor responses and thus should align with users’ embodied expectations.

5.2 Living the Journey

5.2.1 Engagement & Exploration. Beyond navigational affordances, the TrainCafé also engaged sensory, emotional, and narrative dimensions that shaped the overall quality of the experience. Despite the initial surprise, participants described the café environment as *“comfortable,” “cool,” “enjoyable,”* and *“a great place to chill and relax”*. Besides, it offered a visually stimulating and novel atmosphere that sparked *“curiosity”* and the desire to *“look around”* and *“explore the café scenery”*, making the whole train-riding experience feel more *“fancy”* and *“engaging”*. In contrast, the standard environment was described as *“just a normal train,” “familiar and basic”*.

5.2.2 Immersion & Real-World Awareness. Across both conditions, participants reported a sense of presence and realism that made them feel as if they were truly there. However, the café condition was experienced as more immersive and absorbing, whereas the MR see-through window in the baseline condition was described as having *“ruined the immersion”*. Participants acknowledged that the see-through window was *“cool,”* because it felt realistic and *“brought a sense of reality,”* for example by helping them *“understand [their] location in the physical room”*. Yet, this realism also introduced confusion. Instead of seeing a *“moving landscape”* as they would expect from a train window, participants saw *“other people, chairs, and walls”*, which they found distracting and disruptive. Participants suggested that this could be improved by simulating different landscape views through the window or by giving users control over when to activate the mixed reality window, such as when real-world awareness is needed (e.g., preparing to disembark, getting information about ticket control, snack service, or noticing when someone approaches them). As social information was not implemented in the current prototype, participants found it *“strange that there were no other people”*, which made some feel *“lonely”* or even *“scared”*. They expressed that seeing other passengers walking around or engaging in *“people-watching”* would enhance the sense of presence. However, several participants also voiced the need for social control, wanting to know *“whether others are observing or even taking my things while I’m immersed in VR”*, or preferring to focus without the distraction of others. These responses suggest

that social needs in VR are highly individual, pointing towards the potential of adaptive systems that balance social presence and privacy based on user preferences.

5.2.3 Future Applications & Enhancements. Overall, participants expressed strong acceptance towards having a virtual café or other immersive environments during train journeys, mentioning its potential to serve as a space for diverse needs, including immersive entertainment and relaxation space to enjoy breathtaking natural scenes, such as a view of the “Matterhorn”, as a “virtual workspace” that could “help concentrate”, as a private space that could be designed to feel “spacious” and “take away the feeling of being in a tin of sardines”, or to “filter out annoying stuff” and reduce distractions. Participants also offered suggestions to enhance the system by adding atmosphere and sensory depth. They proposed including “subtle train sounds” or “vocal announcements” like in a real train for travel related information, and introducing music or cosy ambient sounds to enrich the experience. Tactile sensations could also be incorporated, such as “feeling the train vibrations”, which would “increase the sense of presence”. Some participants further suggested adding interactive elements in the café or making the environment more dynamic, for example through “animations or moving things in the room” or by “having someone playing piano”. Other participants suggested integrating real-world services, such as “ordering actual coffee or snacks in the virtual café”. Regarding the narrative, one participant noted that “storywise, the café did not really match the travel context well, it was out of expectation.” (P18) They proposed exploring alternative virtual metaphors that better fit the travel theme, such as “having railway crossing barriers” to indicate boundaries instead of using stanchions. In other contexts, such as air travel, participants imagined that “the virtual place could turn into a space simulation, using stars or other planets as metaphors.”

6 DISCUSSION

6.1 Towards Always-On VR in Everyday Life

Although the experiment took place within a controlled simulation, it offers an early lens for understanding always-on pervasive VR/MR, where users navigate and engage in real-world activities while wearing a headset, and how such systems might shape everyday experience and behaviour. This further blurs the boundary between the virtual and the physical world and opens discussions on how extended use of head-mounted displays may merge the real and the virtual in continuous, situated ways. It also highlights the need to design for seamless blending and transitions between realities in everyday life scenarios. These explorations align with ongoing HCI research on ubiquitous virtual and mixed reality [9, 36] and personal spatial computing [38], which envision instant switching between realities and immersion in personalised surroundings that dynamically adapt to users’ needs.

6.2 Risks & Societal Implications

Through the study, we see how such reality-aware, always-on immersive systems can reshape everyday life and change human behaviour. Although 24/7 immersive systems can offer comfort and productivity, they also come with societal risks:

1) *Dual Presence.* Reality-aware systems may normalise a state of continuous partial presence, where people exist simultaneously in both physical and virtual worlds. Even when physically co-located, users may still be “somewhere else” [1]. Such dual existence raises questions about attentiveness, social connectedness, and the erosion of embodied, shared experiences.

2) *Deepening the Enclosed Loop of Living in VR.* By allowing “transitioning and navigating in real life without leaving VR”, designers risk creating an enclosed loop in which users no longer need to remove the headset. This could normalise prolonged engagement in virtual life [10], further detaching individuals from their physical surroundings and reducing serendipitous encounters with others.

3) *Overstimulation Through Continuous Immersion.* It risks overstimulation and dependency. Being “always on” might intensify cognitive fatigue [4, 16], distort time perception, and reduce moments of rest or reflection.

4) *Reality as a Configurable Experience.* Such systems philosophically challenge what “real” means when perception is no longer limited by the physical environment and presence [5, 15], as each user can inhabit their own customised version of reality.

Building on these risks, we propose a set of actionable design guidelines to support designers in developing immersive everyday MR technologies and on-the-go VR experiences, outlined below:

- (1) *Grounding:* maintaining embodied awareness of both the physical and virtual worlds, and supporting attention management by enabling balance between virtual presence, physical presence, and moments of dual presence.
- (2) *Interruption:* designing exit points.
- (3) *Modulation:* incorporating moments of intentional calm and regulate sensory load.
- (4) *Anchoring:* preserving shared perceptual reference points to maintain mutual understanding across individually customised realities.

7 FUTURE WORK & CONCLUSION

Building on this prototype, future work will focus on extending the system toward real-world deployment and integration. Possible directions include: (1) connecting the prototype with live data via the SBB API to provide real-time travel information; (2) integrating other real-world information such as passengers and ticket controllers; (3) ensuring scalability across different train models and vehicle types through dynamic spatial mapping of interior layouts; (4) enabling more customisable environments with diverse themes to enrich the experiential dimension; and (5) conducting real-world studies that go beyond lab settings, involving authentic travel tasks such as finding toilets or restaurants on board.

This paper presents TrainCafé artefact and an empirical study investigating in-vehicle navigation while being immersed in VR. Our findings show that, through virtual metaphors, participants were able to successfully access real-world spatial information without breaking immersion, and that overall travel experiences can be enhanced through VR. We therefore contribute to the broader discourse on reality-aware, always-on mobile virtual reality, where immersive technologies increasingly intertwine with everyday life.

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